

Differential Splicing with Subisoform Covariance

Purpose

The hierarchical model estimates transcript abundance from equivalence-class (EC) counts, but full-length transcripts are often less identifiable than the local splicing choices needed for biological interpretation. The implemented differential-splicing layer therefore collapses transcripts to local splice paths, estimates covariance from the same paired-primer EC likelihood, and tests path usage across biological replicates.

The analysis distinguishes two sources of variation:

1. inferential uncertainty caused by finite UMIs and ambiguous ECs;
2. biological variation among mice with the same condition and cell type.

Inferential covariance is not used as a replacement for biological replication. It determines the precision assigned to each pseudobulk, while a separate variance component represents between-mouse variation.

Subisoforms

Atomic exon annotation

Annotated exon coordinates are read from the GTF as zero-based, half-open intervals. Overlapping exons within each transcript are merged. All exon boundaries in a gene define atomic intervals. An atomic interval is *constitutive* when it is exonic in every annotated spliced transcript of that gene. Adjacent constitutive intervals are merged into constitutive anchor regions.

Splice-path blocks

Each pair of consecutive constitutive anchors defines one block. A transcript’s subisoform in that block is its ordered set of exonic intervals between the anchors. Transcripts with the same local path are collapsed even if their full-length structures differ elsewhere. Blocks with only one path are discarded.

Terminal pseudo-anchors are added before the first and after the last constitutive region. A gene with no shared constitutive region is represented by one whole-gene block. These rules include alternative first and last regions while retaining the local interpretation of internal blocks. Simple two-path blocks correspond to familiar skipped-exon, mutually-exclusive-exon, alternative donor, alternative acceptor, and retained-intron events. The general representation also permits more than two paths and nested events.

Let b index blocks in gene g , and let C_{gb} be its path-by-transcript incidence matrix. Precursor features are retained in the EC likelihood as nuisance features but are not assigned to spliced paths.

Paired EC Likelihood

For one mouse-by-cell-type pseudobulk and primer p , let $\mathbf{c}_p$ be the EC counts, $N_p = \mathbf{1}'\mathbf{c}_p$, and A_p the fixed weighted EC design. The transcript abundance vector $\boldsymbol{\theta}$ gives

$$\mathbf{q}_p = A_p \boldsymbol{\theta}, \quad Z_p = \mathbf{1}'\mathbf{q}_p = \mathbf{d}_p' \boldsymbol{\theta}, \quad \mathbf{d}_p = A_p' \mathbf{1}.$$

Conditioning on N_p ,

$$\mathbf{c}_p \mid N_p, \boldsymbol{\theta} \sim \text{Multinomial} \left(N_p, \frac{\mathbf{q}_p}{Z_p} \right).$$

The oligo(dT) and random-hexamer likelihoods are evaluated separately and their information matrices are added. The weighted designs preserve their different positional and transcript-exposure assumptions.

Only ECs whose compatible transcripts map uniquely to one gene enter that gene's covariance calculation. Cross-gene ECs remain part of the genome-wide hierarchical fit but are excluded from this local curvature approximation.

Transcript Information

Write $\eta_t = \log \theta_t$. Define the EC-specific transcript responsibilities and the primer-level sampled-transcript distribution

$$\mathbf{r}_{ep} = \frac{A_{ep} \odot \boldsymbol{\theta}}{q_{ep}}, \quad \mathbf{r}_{0p} = \frac{\mathbf{d}_p \odot \boldsymbol{\theta}}{Z_p}.$$

The expected Fisher information in transcript-log-abundance coordinates is

$$I_{\eta,p} = N_p \left[\frac{\text{diag}(\boldsymbol{\theta}) A_p' \text{diag}(\mathbf{q}_p^{-1}) A_p \text{diag}(\boldsymbol{\theta})}{Z_p} - \mathbf{r}_{0p} \mathbf{r}_{0p}' \right].$$

Equivalently, this is N_p times the variance across observed ECs of the conditional transcript responsibility. Ambiguous ECs have similar responsibilities under competing transcripts and therefore contribute little information in those contrast directions. The total information is

$$I_\eta = \sum_p I_{\eta,p}.$$

This is the expected-information counterpart of the missing-information identity used to obtain uncertainty after EM.

Path Coordinates and Covariance

For block b , spliced transcript abundances are normalized within their gene and collapsed:

$$\boldsymbol{\psi}_b = C_b \frac{\boldsymbol{\theta}_{\text{spliced}}}{\mathbf{1}' \boldsymbol{\theta}_{\text{spliced}}}.$$

An orthonormal Helmert basis Q_b , satisfying $Q_b' \mathbf{1} = 0$, defines identifiable log-ratio coordinates

$$\mathbf{y}_b = Q_b' \log \boldsymbol{\psi}_b.$$

If $J_b = \partial \mathbf{y}_b / \partial \boldsymbol{\eta}'$, the profile-style delta covariance is

$$V_b \approx J_b I_\eta^+ J_b',$$

where I_η^+ is the inverse on the positive-eigenvalue subspace. The implementation explicitly projects J_b onto the null space of I_η . A block is reported as unidentified rather than assigning zero variance when its path contrast overlaps an information-null direction.

This calculation profiles all transcript directions, including differences among full-length transcripts assigned to the same local path. It can therefore yield finite path covariance even when individual transcripts are not identifiable.

Local Path Refinement

The hierarchical cell decoder supplies a baseline for each pseudobulk. Cell transcript proportions are weighted by their UMI totals and aggregated within mouse, cell type, and condition. For a block, local parameters $\boldsymbol{\delta}_b$ multiply all transcripts carrying the same path:

$$\theta_t(\boldsymbol{\delta}_b) \propto \theta_t^{(0)} \exp\{Q'_{b,s(t)} \boldsymbol{\delta}_b\}.$$

The total spliced mass and the transcript mixture within each path remain fixed. Bounded L-BFGS optimizes the exact paired multinomial likelihood, limiting path perturbations to $|\delta| \leq 20$ to prevent floating-point underflow at a simplex boundary. Under this parameterization,

$$\mathbf{y}_b(\boldsymbol{\delta}_b) = \mathbf{y}_b^{(0)} + \boldsymbol{\delta}_b.$$

The inverse Fisher information for $\boldsymbol{\delta}_b$ gives a second, *conditional* covariance. It is useful for optimization diagnostics and parametric bootstrap validation but is less conservative than the profile covariance because within-path transcript mixtures are fixed.

Finite covariance alone is not sufficient for a Gaussian approximation. Weakly informed path fits can move to a boundary under repeated sampling. The implemented reliability rule requires

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \hat{\psi}_{bis} \geq 0.15,$$

so every retained orthonormal log-ratio direction has effective Fisher information of at least eight and no fitted path lies near the simplex boundary. All fits remain in the estimate output, with separate identifiable and reliable flags; only reliable fits enter the differential test. The thresholds were selected from the label-free parametric bootstrap described below.

Differential-Splicing Test

Let i index independent mouse pseudobulks within a cell type. For each block,

$$\hat{\mathbf{y}}_{bi} = X_i B_b + \boldsymbol{\epsilon}_{bi}, \quad \text{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I.$$

Here V_{bi} is the EC-derived inferential covariance. The scalar τ_b^2 represents biological variation not explained by condition. It is estimated by restricted maximum likelihood under the null design, which excludes the condition coefficients. Generalized least squares combines pseudobulks using $(V_{bi} + \tau_b^2 I)^{-1}$. A multivariate Wald test evaluates all non-reference condition coefficients. Benjamini–Hochberg correction is performed separately for conditional and profile covariance results.

Condition labels are permuted while preserving their observed group sizes. The null REML variance is invariant to these permutations and is reused. Permutation p-values and the distribution of permuted asymptotic p-values are written as calibration diagnostics. The primary p-value is the asymptotic Wald value because a small fixed number of permutations cannot resolve the tail needed for genome-wide false-discovery control.

For a two-path block, the test has one log-ratio coordinate and is equivalent to a precision-weighted differential-PSI analysis. Multi-path blocks receive one omnibus test with $S_b - 1$ coordinates per condition contrast.

Validation

Numerical tests cover:

- construction of a skipped-exon block from a synthetic GTF;
- finite path covariance when two indistinguishable transcripts collapse to one path;
- rejection of a path contrast when all paths produce the same EC;
- equality of the analytic two-path Fisher variance and the binomial result;
- agreement between analytic variance and parametric multinomial simulations;
- recovery of an imposed multivariate condition effect by GLS.

The production validation additionally simulates EC counts from fitted pseudobulk path models, refits the local path parameters, and compares empirical squared error with analytic conditional variance. It reports component-wise squared standardized error and 95% interval coverage for each validation fit. Condition-label permutations provide a null calibration check for the complete differential-testing layer.

In a two-path smoke analysis, all 1,257 eligible pseudobulk fits converged. Conditional covariance was identifiable for 95.5% of fits and profile covariance for 52.3%. Before reliability filtering, fits with information below eight and paths below 15% usage showed boundary-driven mode switching. Above both thresholds, mean squared standardized error was 1.09 and component-wise 95% coverage was 94.0%. Independent normal simulations of the REML–GLS implementation gave type-I error between 2.7% and 5.0% at a nominal 5% across zero, moderate, and large biological-variance settings.

The genome-wide microglialess analysis considered blocks with up to eight paths. After the gene-count filter, 2,820 blocks contributed 46,393 pseudobulk estimates; every local optimization converged. Conditional covariance was identifiable for 73.5% of estimates and 12.0% passed both reliability criteria. Profile covariance was identifiable for 29.8% and 7.9% passed both criteria. The retained estimates produced 170 conditional and 113 profile differential tests. Five conditional and three profile tests had unadjusted $p < 0.05$, but none passed 5% false-discovery control.

Across 3,000 production bootstrap refits, mean squared standardized error was 1.52 and 95% interval coverage was 94.6%. Among permuted-label tests, 2.9% of conditional and 2.5% of profile asymptotic p-values were below 0.05. Thus the implemented filter gives slightly conservative null calibration. The lack of discoveries reflects both this conservatism and the small number of mouse replicates available for many cell-type–condition combinations.

Coverage-cutoff sensitivity

The minimum gene-assigned UMI count per pseudobulk was varied while retaining the same covariance and path-proportion reliability rules:

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

The 500-UMI permutation estimate is based on only 100 null tests and is therefore noisy. No cutoff produced a discovery at 5% FDR.

Asymptotic power was also calculated for reliable two-condition, two-path tests using their fitted inferential covariance and REML biological variance. An ILR effect of 0.5, equivalent to a twofold path-odds change, gave mean conditional-test power of 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The apparently higher power at 200 and 500 UMIs is selected from only eight and two eligible tests. Among 103 test identities shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449, respectively. Thus lower cutoffs do not degrade calibrated per-test precision after reliability filtering; they mainly increase the number of testable blocks. A 25-UMI coverage cutoff is the preferred default for this dataset.

Limitations

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could shrink path-specific biological covariances across genes.